GC University Faisalabad

Department of Information Technology

Assignment No. 1

Course: Information Security (IS)

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Question No. 1

Part (a): Exhaustive Key Search

What it is

An exhaustive key search, also called a brute-force attack, is the most basic way of breaking an encryption system. The attacker already has some ciphertext and does not know the key, so he simply tries every possible key one after another. For each key he decrypts the ciphertext and looks at the result. When the output turns into readable text (or matches a piece of known plaintext), that key is the correct one. Nothing clever about the cipher is used here; it only needs time and computing power.

The work grows very fast with the key length. A key of n bits has 2^n possibilities, and on average the attacker finds the right key after trying about half of them. This is why the key length is the first thing people check when they judge how strong a cipher is.

Where it is used and why

- **Breaking old or weak ciphers:** When the cipher itself is weak or old and has a short key, for example DES with its 56-bit key. Here the key space is small enough for a brute-force attempt to actually finish.
- **Password and PIN recovery:** When all that is known is that a file or password is protected, and no shortcut exists. Recovery tools try all short passwords or PINs because humans often pick short ones.
- **Measuring the strength of a cipher:** Security researchers and designers run it (in theory) as a yardstick. If no attack is faster than exhaustive search, the cipher is considered strong for its key size.
- **Forensics and data recovery:** Digital forensic teams sometimes have to open an encrypted drive or archive of an old system when the owner is not available or does not cooperate.
- **When nothing better is available:** Exhaustive search is also the fall-back when the attacker has a lot of resources, such as clusters of GPUs or special hardware like FPGAs, and the cipher has no known weakness.

The reason it is used in all these cases is simple: it works against every cipher, it is easy to program, and it can be split across many machines because each machine can take a different part of the key space.

Example

Take a Caesar cipher. The key is just a shift from 1 to 25, so there are only 25 possible keys. Suppose we intercept the ciphertext "KHOOR". We try each shift in turn:

Key (shift)	Result of decryption
1	JGNNQ
2	IFMMP
3	HELLO <- readable
4	GDKKN

At shift 3 the text becomes HELLO, which is a real word, so the key is 3. It took only three tries here. The same idea against a modern cipher is not realistic, as the table below shows (assuming a very strong attacker that tests 10^12 keys every second).

Key size	Number of keys	Approximate time to try all
56 bits (DES)	7.2 x 10^16	about 20 hours
128 bits (AES-128)	3.4 x 10^38	about 10^19 years
256 bits (AES-256)	1.2 x 10^77	far beyond the age of the universe

So exhaustive search is practical only when the key space is small. For AES with 128 bits or more it is not a threat.

Part (b): Cyber Security and Why It Matters

Definition

Cyber security means protecting computers, networks, servers, mobile devices, applications and the data stored in them from attacks, damage or unauthorised access. It covers technology, processes and also the people who use the systems. The main goals are usually described as the CIA triad: confidentiality (only the right people can read the data), integrity (the data is not changed without permission) and availability (the system and data can be used whenever needed).

Why it is important

- **Society depends on digital systems.** Banks, hospitals, schools, power plants and government offices all run on computers now. An attack on them can stop daily life.
- **Protection of sensitive data.** Personal details, medical records, card numbers and company secrets are valuable, and criminals sell them. A leak can lead to identity theft and fraud.
- **Financial and reputational loss.** A successful ransomware or data breach costs money through downtime, recovery, legal fines and lost customers. Reputation is damaged as well.
- **National security.** Attacks on critical infrastructure and state-backed hacking groups are a threat to national security.
- **Trust in online services.** Online banking, e-commerce and cloud services only work when users trust that their transactions are safe.
- **Legal and regulatory compliance.** Many countries have laws on data protection, so organisations must show that they take security seriously.

Current research challenges

- **AI-driven attacks and defence.** Attackers now use machine learning to write convincing phishing mails, make deepfake voices and videos, and find vulnerabilities faster. Defenders also use AI, but AI models themselves can be fooled by adversarial inputs or poisoned training data.
- **Internet of Things (IoT) security.** Billions of cheap smart devices (cameras, sensors, wearables) have weak passwords, limited memory and rarely get updates. Designing lightweight security for them is still an open problem.
- **Quantum threat to cryptography.** A large enough quantum computer could break RSA and elliptic-curve cryptography. Researchers are working on post-quantum algorithms, and moving old systems to them will take years.
- **Cloud and data security.** Misconfigured storage, shared responsibility between provider and customer, and multi-cloud setups make it hard to keep a clear view of what is protected.
- **Software supply-chain attacks.** Attackers target a trusted supplier or software update to reach many victims at once, as seen in the SolarWinds incident.
- **Zero-day vulnerabilities.** Vulnerabilities that the vendor does not know about cannot be patched, and detecting such attacks is difficult.
- **Privacy versus security.** Collecting more data helps detect threats, but it can harm user privacy. Finding the right balance is difficult.
- **Human factor and skills gap.** Most breaches begin with a human mistake, such as clicking a bad link. Training people is slow, and there is also a shortage of skilled security professionals.

Part (c): Important Terms

Term	Description	Example
Worm	A worm is a self-replicating program that spreads by itself through networks, usually by exploiting a vulnerability, and does not need a host file or any action from the user. Because of this it can infect thousands of machines in a short time and slow down networks.	The Morris worm (1988) and the WannaCry worm (2017), which spread through a Windows file-sharing flaw.
Spyware	Spyware secretly gets installed on a device and collects information such as browsing habits,	Keyloggers that record everything typed, and the

Term	Description	Example
	passwords or keystrokes, then sends it to the attacker without the user knowing.	Pegasus spyware that targeted phones.
Virus	A virus attaches itself to a legitimate file or program. It runs only when that file is opened and then copies itself into other files. It needs a human action (such as running the infected file) to spread.	The ILOVEYOU virus (2000) that came as an email attachment, and the CIH (Chernobyl) virus.
Trojan	A Trojan pretends to be useful software but hides a harmful function inside. It does not replicate itself; the user is tricked into installing it, and it may then open a back door for the attacker.	A fake free game or cracked software that installs a remote access tool; banking Trojans like Zeus.
Adware	Adware displays unwanted advertisements, pop-ups or redirects, often bundled with free software. It may also track the user's browsing to target ads and can slow the device down.	Browser toolbars installed along with free downloads; the Fireball adware.
Ransomware	Ransomware encrypts the files of the victim (or locks the whole system) and demands payment, usually in cryptocurrency, in return for the decryption key.	CryptoLocker, WannaCry, and the attack on Colonial Pipeline (2021).
Active Cyber Attack	In an active attack the attacker does something to the system or the data in transit: alters, injects, blocks or destroys it. Such attacks are easier to detect because the system is affected, but harder to prevent completely.	Masquerading as another user, replaying a captured login message, modifying a message, and denial-of-service (DoS).
Passive Cyber Attack	In a passive attack the attacker only watches or listens and does not change anything, so the victim usually does not notice it. The aim is to learn information. Encryption is the best defence, as prevention is more useful than detection here.	Eavesdropping on unencrypted Wi-Fi traffic, sniffing packets, and traffic analysis (studying who talks to whom and how often).
Information Assurance	Information assurance (IA) is the practice of making sure that information is reliable and safe throughout its life. It goes beyond protection from hackers: it includes managing risk, backups, policies, and keeping the five properties of confidentiality, integrity, availability, authentication and non-repudiation.	A hospital that encrypts patient records, keeps backups, controls who can read them and keeps logs for audit.

Note: the quick difference between a virus and a worm is that a virus needs a host file and human action, while a worm spreads on its own. A Trojan, on the other hand, depends on deceiving the user.

Question No. 2: Secret Key Cryptography (AES)

Part (a): History of AES

By the 1990s the Data Encryption Standard (DES), adopted in 1977 with a 56-bit key, had become too weak. In 1998 a machine built by the Electronic Frontier Foundation broke a DES key in a little over two days, and Triple DES, the temporary fix, was slow in software. So the U.S. National Institute of Standards and Technology (NIST) decided to choose a new standard.

- **1997:** In January 1997 NIST announced a public competition for the Advanced Encryption Standard. The requirements were a 128-bit block and key sizes of 128, 192 and 256 bits, with good security, speed and low memory use, and royalty-free use.
- **1998:** Fifteen candidate algorithms from different countries were accepted.

- **1999:** Five finalists were announced: MARS, RC6, Rijndael, Serpent and Twofish.
- **2000:** On 2 October 2000 NIST selected Rijndael, designed by two Belgian cryptographers, Joan Daemen and Vincent Rijmen, as the winner. It was chosen for its good balance of security, speed and simplicity.
- **2001:** AES was published as FIPS PUB 197 in November 2001 (effective May 2002).
- **2003 onwards:** The U.S. government approved AES for protecting classified information, with 192 and 256-bit keys for top secret. Since then it has become the most widely used symmetric cipher in the world (Wi-Fi WPA2, HTTPS/TLS, disk encryption, VPNs).

Note that AES is not exactly the same as Rijndael. Rijndael supports block sizes of 128, 192 and 256 bits, while AES fixes the block size at 128 bits.

Part (b): Working Flow Chart of AES

AES is a symmetric block cipher. It works on a 128-bit block of data, which is arranged as a 4 x 4 matrix of bytes called the state (filled column by column). The number of rounds depends on the key length: 10 rounds for a 128-bit key, 12 for 192 and 14 for 256. Unlike DES, AES does not use a Feistel structure; it is a substitution-permutation network where the whole block is changed in every round.

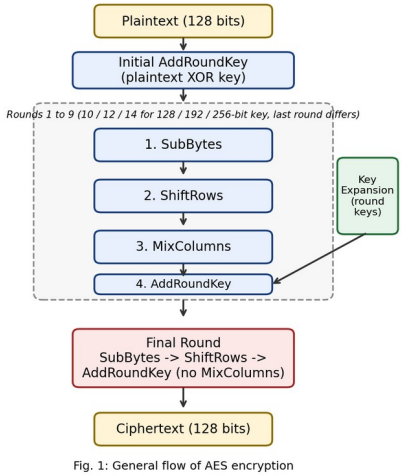


Fig. 1: General flow of AES encryption

Steps in short

- **Key expansion:** The 128-bit key is expanded into separate round keys (11 round keys for AES-128).
- **Initial round:** The plaintext block is XORed with the first round key.
- **Main rounds (9 for AES-128):** Each of these rounds applies SubBytes, ShiftRows, MixColumns and AddRoundKey.
- **Final round:** Same as a main round but without MixColumns.

What each transformation does:

- **SubBytes:** every byte is replaced using a fixed 16 x 16 lookup table (the S-box). This gives confusion, since the relation between key and ciphertext becomes non-linear.
- **ShiftRows:** row 0 stays as is, row 1 moves one byte to the left, row 2 two bytes, and row 3 three bytes (cyclic shift).
- **MixColumns:** each column is multiplied with a fixed matrix in the field GF(2^8). This mixes the four bytes of the column and gives diffusion.
- **AddRoundKey:** the state is XORed with the round key for that round, which is where the secret key actually enters.

Decryption uses the inverse of these operations (InvSubBytes, InvShiftRows, InvMixColumns) with the round keys applied in reverse order.

Part (c): One Round of AES-128, Step by Step

The values below are the ones given in the question. In the matrices, the bytes are filled column by column, which is the AES convention. All numbers are in hexadecimal.

Item	Value
Plaintext	32 43 f6 a8 88 5a 30 8d 31 31 98 a2 e0 37 07 34
Key	2b 7e 15 16 28 ae d2 a6 ab f7 15 88 09 cf 4f 3c

Step 0: Plaintext and key as state matrices

Plaintext (state)

32	88	31	e0
43	5a	31	37
f6	30	98	07
a8	8d	a2	34

Bytes in order: 32 43 f6 a8 88 5a 30 8d 31 31 98 a2 e0 37 07 34

Cipher key (round key 0)

2b	28	ab	09
7e	ae	f7	cf
15	d2	15	4f
16	a6	88	3c

Bytes in order: 2b 7e 15 16 28 ae d2 a6 ab f7 15 88 09 cf 4f 3c

Step 1: Initial AddRoundKey (state XOR round key 0)

Each byte of the plaintext is XORed with the byte at the same place in the key. For example, the first byte is 32 XOR 2b = 19, and the second is 43 XOR 7e = 3d.

Result after initial AddRoundKey (input to round 1)

19	a0	9a	e9
3d	f4	c6	f8
e3	e2	8d	48
be	2b	2a	08

Bytes in order: 19 3d e3 be a0 f4 e2 2b 9a c6 8d 2a e9 f8 48 08

Now round 1 begins with the four transformations asked in the question.

Step 2: SubBytes

Every byte is replaced from the S-box. The high nibble selects the row and the low nibble the column. Some examples: 19 -> row 1, column 9 -> d4; 3d -> row 3, column d -> 27; e3 -> 11; be -> ae; a0 -> e0; f4 -> bf.

State after SubBytes

d4	e0	b8	1e
27	bf	b4	41
11	98	5d	52
ae	f1	e5	30

Bytes in order: d4 27 11 ae e0 bf 98 f1 b8 b4 5d e5 1e 41 52 30

Step 3: ShiftRows

Row 0 is not shifted. Row 1 is rotated left by 1, row 2 by 2 and row 3 by 3 positions. For example, row 1 (27 bf b4 41) becomes (bf b4 41 27) and row 3 (ae f1 e5 30) becomes (30 ae f1 e5).

State after ShiftRows

d4	e0	b8	1e
bf	b4	41	27
5d	52	11	98
30	ae	f1	e5

Bytes in order: d4 bf 5d 30 e0 b4 52 ae b8 41 11 f1 1e 27 98 e5

Step 4: MixColumns

Each column is multiplied by the fixed matrix [02 03 01 01; 01 02 03 01; 01 01 02 03; 03 01 01 02] over GF(2^8), with the reduction polynomial x^8 + x^4 + x^3 + x + 1 (1b in hex). Addition is XOR. Working of the first byte of column 1 (d4, bf, 5d, 30):

- 02 x d4 = b3 (shift left gives 1a8, which is reduced by XOR with 11b)
- 03 x bf = (02 x bf) XOR bf = 65 XOR bf = da
- so b3 XOR da XOR 5d XOR 30 = 04

The other bytes are found in the same way, which gives:

Column	Input	Output
Column 1	d4 bf 5d 30	04 66 81 e5
Column 2	e0 b4 52 ae	e0 cb 19 9a
Column 3	b8 41 11 f1	48 f8 d3 7a
Column 4	1e 27 98 e5	28 06 26 4c

State after MixColumns

04	e0	48	28
66	cb	f8	06
81	19	d3	26
e5	9a	7a	4c

Bytes in order: 04 66 81 e5 e0 cb 19 9a 48 f8 d3 7a 28 06 26 4c

Step 5: Round key 1 (key expansion)

Before the last transformation we need round key 1. The last word of the original key is w3 = 09 cf 4f 3c. We rotate it left (cf 4f 3c 09), pass it through the S-box (8a 84 eb 01) and XOR with the round constant Rcon[1] = 01 00 00 00, which gives 8b 84 eb 01. Then w4 = w0 XOR 8b 84 eb 01 = a0 fa fe 17. The next words are w5 = w1 XOR w4, w6 = w2 XOR w5 and w7 = w3 XOR w6.

Round key 1

a0	88	23	2a
fa	54	a3	6c
fe	2c	39	76
17	b1	39	05

Bytes in order: a0 fa fe 17 88 54 2c b1 23 a3 39 39 2a 6c 76 05

Step 6: AddRoundKey (state XOR round key 1)

The state after MixColumns is XORed with round key 1. Example: 04 XOR a0 = a4, 66 XOR fa = 9c, 81 XOR fe = 7f, e5 XOR 17 = f2.

State at the end of round 1

a4	68	6b	02
9c	9f	5b	6a

7f	35	ea	50
f2	2b	43	49

Bytes in order: a4 9c 7f f2 68 9f 35 2b 6b 5b ea 43 02 6a 50 49

Summary of intermediate results

Stage	State (16 bytes)
Plaintext	32 43 f6 a8 88 5a 30 8d 31 31 98 a2 e0 37 07 34
After initial AddRoundKey	19 3d e3 be a0 f4 e2 2b 9a c6 8d 2a e9 f8 48 08
After SubBytes	d4 27 11 ae e0 bf 98 f1 b8 b4 5d e5 1e 41 52 30
After ShiftRows	d4 bf 5d 30 e0 b4 52 ae b8 41 11 f1 1e 27 98 e5
After MixColumns	04 66 81 e5 e0 cb 19 9a 48 f8 d3 7a 28 06 26 4c
Round key 1	a0 fa fe 17 88 54 2c b1 23 a3 39 39 2a 6c 76 05
After AddRoundKey (end of round 1)	a4 9c 7f f2 68 9f 35 2b 6b 5b ea 43 02 6a 50 49

These results match the worked example of AES-128 given in the official standard FIPS 197 (Appendix B), so the calculation is consistent with the reference.

Part (d): Comparison of DES, 3DES and AES

All three are symmetric (secret key) block ciphers, but they come from different periods and have different strength.

Feature	DES	3DES	AES
Year	1977	1998 (as a standard, ANSI X9.52)	2001
Designer	IBM (Feistel), adopted by NIST	Based on DES, applied three times	Daemen and Rijmen (Rijndael)
Structure	Feistel network	Feistel (three DES operations: E-D-E)	Substitution-permutation network
Block size	64 bits	64 bits	128 bits
Key size	56 bits (64 with parity)	112 or 168 bits (3 keys)	128, 192 or 256 bits
Rounds	16	48 (3 x 16)	10, 12 or 14
Effective security	Weak, broken by brute force	About 112 bits at most (meet-in-the-middle attack)	128 / 192 / 256 bits
Speed	Fast in hardware, moderate in software	About 3 times slower than DES	Fast in both hardware and software
Known attacks	Brute force, linear and differential cryptanalysis	Sweet32 (small 64-bit block)	No practical attack on full rounds
Current status	Withdrawn, obsolete	Deprecated; NIST disallowed it for new use after 2023	Current standard, widely used

In conclusion, DES is too short-keyed for today, 3DES was only a stop-gap that is slow and has a small block, and AES is the best of the three in security and performance, so it is what new systems should use.

Part (e): Traffic Confidentiality and Placement of Encryption Function

Traffic confidentiality

Even when the content of messages is encrypted, an attacker who is watching the network can still learn a lot from the traffic itself: who is communicating with whom, how often, how long the messages are, and at what

time. This is called traffic analysis. For example, a sudden rise in messages between two offices may show that something important is going on, even if nothing can be read. Traffic confidentiality means hiding these patterns. The information an attacker can try to get includes the identity of the partners, the frequency and length of messages, and any change in the pattern that matches a special event.

Common countermeasures are link encryption (which also hides the headers), traffic padding (sending dummy data so that the volume looks constant), and routing through intermediate nodes so that real source and destination are hidden.

Placement of encryption function

If encryption is to be used on a network, we must decide where in the path to put it. There are two main choices.

- **Link encryption:** Encryption devices are placed at both ends of every communication link (between each pair of nodes). All the traffic on the link, including the headers, is encrypted. But each packet must be decrypted at every router or switch to read the address and then encrypted again for the next link. So the data is in plaintext inside each intermediate node, and those nodes must be trusted. It also needs many devices and keys (one pair for each link).
- **End-to-end encryption:** Encryption is done at the source and the data is decrypted only at the destination. The user data is encrypted, but the headers must stay in plaintext so that the network can route the packet. The data stays encrypted all the way, even in the middle nodes, but an attacker can still see the headers and so do traffic analysis.

Point	Link encryption	End-to-end encryption
Where applied	On each link between nodes	At source and destination only
Headers	Encrypted (hidden)	Sent in plain so network can route
Data inside nodes	Plaintext at each node	Remains encrypted
Traffic analysis	Protected on the link	Exposed (headers visible)
Number of keys	Many (one per link)	One per pair of users / hosts
Who controls it	Network provider	The end users

In practice, both are often used together: end-to-end encryption protects the data from source to destination, and link encryption protects the headers on the links where traffic analysis is a risk. Encryption may also be placed at different layers, such as the network layer (IPsec) or the application layer (for example, secure email), and the choice depends on what must be protected and who can be trusted.

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